A PROJECT TITLE

ON

Player Performance Prediction Using Ridge and Lasso Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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1. Abstract:

Machine Learning has become an essential technology for analyzing large amounts of sports data and predicting player performance with improved accuracy. Accurate performance prediction helps coaches, selectors, sports analysts, and management make informed decisions regarding team selection, player development, and match strategies. The objective of this project is to develop a machine learning model that predicts player performance using Ridge Regression and Lasso Regression techniques.

The project utilizes historical player statistics such as matches played, innings, runs scored, batting average, strike rate, wickets, economy rate, catches, age, and experience as input features. The dataset is collected from publicly available sports databases and is preprocessed by handling missing values, removing duplicate records, encoding categorical variables, and scaling numerical features. Exploratory Data Analysis (EDA) is performed to understand the relationships among different attributes and identify the most influential factors affecting player performance.

Two regularized regression algorithms, Ridge Regression and Lasso Regression, are implemented and trained using the processed dataset. Ridge Regression minimizes overfitting by applying L2 regularization, while Lasso Regression performs both regularization and feature selection using L1 regularization. The models are evaluated using performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Square Error (RMSE), and R^2 Score to determine the better-performing model.

The developed system provides reliable predictions of player performance based on historical data, assisting sports professionals in making data-driven decisions. The project also demonstrates the effectiveness of regularization techniques in improving prediction accuracy and handling multicollinearity. In the future, the system can be enhanced by incorporating advanced machine learning algorithms such as Random Forest, XGBoost, and Neural Networks, as well as integrating real-time sports data for more accurate and dynamic predictions.

2. INTRODUCTION:

Sports analytics has experienced significant growth with the advancement of Machine Learning and Artificial Intelligence. Predicting a player's future performance based on previous statistics enables coaches and team management to make objective decisions instead of relying solely on experience or intuition. Machine Learning algorithms can identify hidden patterns within historical data and generate accurate predictions that support player selection and strategic planning.

This project focuses on predicting player performance using Ridge Regression and Lasso Regression, which are regularized versions of Linear Regression. These algorithms effectively reduce overfitting and improve model generalization, especially when datasets contain correlated features. The project involves data collection, preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and comparison. The final outcome is a predictive model capable of estimating player performance with high

accuracy while identifying the most influential performance factors.

3. SOFTWARE REQUIREMENTS:

- Operating System: Windows 10/11
- Programming Language: Python 3.x
- IDE: Jupyter Notebook / Google Colab / Visual Studio Code
- Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
- Machine Learning Framework: Scikit-learn
- Dataset Source: Kaggle Player Performance Dataset

Signature of the Guide **Course Instructor**